Fiscal Year:	FY 2005	**Task Last Updated:**	FY 01/10/2005
PI Name:	Hart, John E. Ph.D.		
Project Title:	Stratified Taylor-Couette Flow with Radial Buoyancy		
Division Name:	Physical Sciences		
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Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Dynamics and stability		
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PI Email:	hart@tack.Colorado.EDU	**Fax:**	FY 303-492-4248
PI Organization Type:	UNIVERSITY	**Phone:**	303-492-8568
Organization Name:	University of Colorado, Boulder		
PI Address 1:	Campus Box 311, Duane Physics Room D318		
PI Address 2:			
PI Web Page:	http://nimbus.colorado.edu/		
City:	Boulder	**State:**	CO
Zip Code:	80309	**Congressional District:**	2
Comments:			
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Contact Monitor:	Hasan, Mohammad	**Contact Phone:**	216-977-7494
Contact Email:	Mohammad.M.Hasan@nasa.gov		
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Tagg, Randy (Univ. Colorado at Denver) Weidman, Patrick (Univ. Colorado at Boulder) Ohlsen, Daniel (Univ. Colorado at Boulder)		
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Task Description:	We are constructing a ground-based experiment on a radial-gravity Taylor-Couette system. The radial gravity is generated by using a magnetic fluid in the working section, and by subjecting the system to a large magnetic field that has axial and azimuthal symmetry in field strength, and which has a large radial gradient in its magnitude. In such a circumstance the magnetohydrodynamic body force M.grad (H) will have a radial component that is linear in the fluid temperature. Thus this new system is identical to one with radial gravity and the associated thermal buoyancy forces that act in the radial direction. It appears possible to build a large Taylor number (i.e. a large Reynolds number) apparatus that has a radial gravitational acceleration of at least 0.3g. Such an experimental facility will be unique for studying stratified shear flow turbulence and sheared convection. The proposed theoretical and ground-based laboratory work will provide the background for a flight experiment design that will take advantage of the microgravity setting to enable experimentation with purely radial gravity fields. The key tasks to be completed in the ground-based study are: (1) Magnet design. Can a permanent magnet of desired symmetry and strength be built? Can we attain the goal (based on reaching really interesting flow regimes) of obtaining Taylor numbers as large as 10^6 with the Richardson number at the same time ranging between something like -5 and +5. (2) Use theory and computation to investigate the relative role of magnetic viscosity and magnetic buoyancy. Non-Newtonian effects that may generate anomalous flows that will interfere with those affected by magnetic gravity. (3) Perform a complete stability analysis. Previous studies have been limited, and have addressed only parts of the linear stability problem for Taylor-Couette motion of a radically stratified ferrofluid. We shall look at axisymmetric and non-axisymmetric modes in a cell with radial magnetic buoyancy and magnetic rotational viscosity, with both cylinders rotating. (4) Numerical simulation. An axisymmetric code will be used to study the behavior of the model cell. What are the end-wall effects? What will these flows look like when Earth's gravity in a direction perpendicular to the cylinder axis is added? (5) Test-bed Experiment. A cell will be constructed with the largest practical magnetic gravity. This will be used to test visualization methods and to compare laboratory results with the computational simulations and theoretical models.6) Based on the result from the above work, a preliminary design for a flight cell will be generated. To perform research a Taylor-Couette (double rotating cylinder) experiment with radial gravity. In this research we shall conduct theoretical analyses of the method of producing radial gravity, of competing and possibly conflicting flows, and of expected instabilities and transition to turbulence in this system. We shall also develop multiple means of visualizing fluid motion in the opaque ferro-fluid that we propose to use. Theory will be tested by physical experimentation and computational simulation. The Taylor-Couette system, in which fluid is contained between two differentially rotating cylinders, is the classic experiment of fluid mechanics. Because the arguments used to interpret Taylor-Couette flows apply to many problems in linear and nonlinear dynamics, this model has become a paradigm. In spite of the long history of the subject, there continues to be intense interest in a wide variety of modifications to the basic system, and hundreds of papers on Taylor vortices and related subjects appear in the literature each year. Even with all this progress, one exceedingly important variation on the classic problem has received very little attention. This concerns the behavior of fluids contained between cylinders rotating at different rates, W1 and W2, in which there is radial gravity, along with a basic radial density stratification. This basic stratification may be either statically stable or statically unstable. Such a system is of fundamental interest because it permits study of such important problems as the nature of the interaction between gravity waves and traveling Taylor-vortex modes. It is also of practical value because for the first time it will allow experimental study of stably and unstably stratified shear flows in re-entrant geometry that are important in many engineering and natural settings. Our studies will clarify the role of magnetic viscosity, vs. that of magnetic buoyancy, in such systems. We will provide theory and computational simulations of a microgravity stratified wind tunnel experiment, and will verify the theory and computations by comparing with terrestrial experiments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Theoretical work was completed that concluded the following: 1) Magnetic viscosity effects are small and therefore negligible in the magnetic TC experiments. Using ferrofluid to simulate Newtonian fluid dynamics in geophysical setting with rotation and stratification is viable. 2) Ferrofluid flows can be measured successfully with ultrasonic velocimetry provided seeing particles are used and the flow rates are of the order 1cm/sec or greater. 3) Anomalous flows due to residual variations of magnetic gravity along the axis of the cylinder (arising for incomplete cancellation of z-periodic vectors in the formation of the dot product of the magnetic field strength), are significant and undesirable. A theory for these shows that the anomalous flows can have a very large effect on the stability of Taylor-Couette motions with strong magnetic forces. Weaker an more uniform magnetic fields must thus be used if we are to simulate geophysical motions. The weakness constraint means that microgravity must be used. However, the uniformity constrain will be difficult to supply, and typically will require experiments in a very shallow layer (in radius). A shallow layer will be dominated by the rigid boundaries (a non-geophysical ingredient). 4) Linear stability theory for axially uniform magnetic fields and terrestrial gravity suggest strong changes in the stability characteristics of TC flow due to magnetic forces. Preliminary experiments in the rotating TC cell support some of the conclusions of the theory, namely the presence of axial columnar convective modes (whereas in normal TC flow the dominant instabilities are axisymmetric). 5) There does appear to be some promise for an ocean model experiment based on the ideas studied in this research. In the ocean model, a horizontal layer of fluid is constrained by the magnets. The spatial variation of magnetic gravity, that was problematic in the TC cell, serves to generate a non-uniform body force that excites the model ocean flow, similar to the way topographic shallowing in the presence of uniform surface heat fluxes causes buoyancy to pile up in certain regions and generate circulations.
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